

CE 310P: Strength of Materials and Structures Lab

Course Code	CE 310P
Course Name	Strength of Materials and Structures Lab
L-T-P-C	0-0-2-1
Prerequisites	CE310: Strength of Materials and Structures / Equivalent
Intended for	UG
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

The curriculum is split into eleven experiments encompassing different structural engineering analysis and strength testing problems typically encountered in the field practices. Validation of the concept through computer programming will also be made as part of this course. The duration of each lab session shall be 3 Lectures

1. Study of stress-strain characteristics of civil engineering materials.
2. Study of reactions in beams with different support conditions.
3. Study of variation of bending moment and shearing force in a beam subjected to various loading conditions.
4. Study of load-deflection characteristics of trusses.
5. Study of load-deflection characteristics of rectangular portal frames with uniform and nonuniform sections.
6. Study of stress and strains in the members of pin-jointed frames.
7. Study of plastic bending of portal frames.
8. Load-deflection study of pinned arches.
9. Study of buckling characteristics of struts.
10. Study of horizontal and vertical deflections of asymmetric sections at various angles and loads.
11. Study of behaviour of circular section under torsion.

Text Books

1. Hibbeler, RC, **Structural Analysis**, 9th Edition, Pearson Education, 2017.
2. Hibbeler, RC, **Mechanics of Materials**, Pearson Education, 10th Edition, 2016.

References

1. Timoshenko, S.P., and Young, D.H., **Elements of Strength of Materials**, 5th Edition, 2003.
2. Beer, F.P., Johnston, E.R., DeWalt, J., and Mazurek, D.F.,
3. Egor P. Popov, **Engineering Mechanics of Solids**, 2nd Edition Pearson Education Inc, 2009
4. Gere and Timoshenko, **Mechanics of Materials**, 2nd Edition, CBS Publishers, 2004.
5. Reddy, CS, **Basic Structural Analysis**, Tata McGraw Hill, 2001.
6. Menon, D., **Structural Analysis**, Narosa Publishing House, 2008.
7. Ross, C.T.F, Case, J, and Chivler, L, **Strength of Materials and Structures**, 4th edition, Butterworth Heinemann, 1999.
8. Megson, THG, **Structural and Stress Analysis**, Butterworth-Heinemann, 2014.